

TIAN-AO (TEO) REN

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RESEARCH INTEREST

My research lies at the intersection of robotics, haptics, and artificial intelligence for intelligent physical interaction in medicine. I am interested in developing multimodal tactile sensing systems, ultrasound-compatible interfaces, and AI-driven contact modeling frameworks that enable robots to perceive, model, and augment expert human touch. More broadly, I seek to bridge hardware innovation, computational learning, and clinical insight to advance human-robot collaboration, skill transfer, and intelligent robotic assistance in healthcare.

EDUCATION

Ph.D. Candidate in Mechanical Engineering (Robotics & Design) Jan 2025 - **Present**
Stanford University California, U.S.

- Advisor: Prof. Mark Cutkosky, PI of [Biomimetics and Dextrous Manipulation Lab\(BDML\)](#)

M.S. in Mechanical Engineering Sep 2023 - Jun 2025
Stanford University California, U.S.

- GPA: **4.0/4.0**
- Depth in **Design Methodology**
- Advisors: Prof. Allison Okamura, PI of [Collaborative Haptics and Robotics in Medicine Lab\(CharmLab\)](#)
Prof. Mark Cutkosky, PI of [Biomimetics and Dextrous Manipulation Lab\(BDML\)](#)

B.Eng. in Robotics Engineering Sep 2019 - Jun 2023
Beijing University of Chemical Technology Beijing, China

- Overall GPA: **90.8/100**, Major GPA: **94.6/100**
- **Outstanding Graduates in Beijing, National Scholarship of China**, Liwen & Yangyan High Scholarship, Jinfa Science & Technology Scholarship, 1st Scholarship.

PUBLICATION *INDICATE EQUAL CONTRIBUTION

Journal Articles

- J. Lai*, **T.-A. Ren***, P. Ye, J. Sun, H. Ren, "**Gravity-Aware Proactive Joint-Level Compensation for Portable Soft Slender Robots Using a Single IMU and Real-Time Simulation**", 2026. *International Journal of Robotics Research (IJRR)*. [\[DOI\]](#)
- E. Rennich, **T.-A. Ren**, J. Kim, J. Di, T. Chen, M. Cutkosky, "**Gentle Grasping with Gecko-Inspired Adhesives in Extreme Environments**", 2025. *IEEE Transactions on Field Robotics (T-FR)*. [\[DOI\]](#)
- J. Lai, Y. Liu, **T.-A. Ren**, Y. Ma, T. Zhang, J. Y. Teoh, M. Cutkosky, H. Ren, "**Single twistable tendon-driven continuum robots**", 2025. *Second round review by Nature Communications*.
- J. Lai*, **T.-A. Ren***, W. Yue, S. Su, J. Y. K. Chan, and H. Ren, "**Sim-to-Real Transfer of Soft Robotic Navigation Strategies That Learns from the Virtual Eye-in-Hand Vision**", 2023. *IEEE Transactions on Industrial Informatics (T-II)*. [\[DOI\]](#) [\[Supply Video\]](#)
- Wang G, **T.-A. Ren**, J. Lai, L. Bai, H. Ren, "**Domain Adaptive Sim-to-Real Segmentation of Oropharyngeal Organs**", 2023. *Medical & Biological Engineering & Computing (MBEC)*. [\[DOI\]](#)

Conference Papers

- **T.-A. Ren**, J. Garcia, S. Hong, J. Grinberg, H. Choi, J. Di, H. Li, D. Grinberg, M. Cutkosky, "**Multimodal Sensing for Robot-Assisted Sub-Tissue Feature Detection in Physiotherapy Palpation**", *Design of Medical Device Conference (DMD)*, 2026. [\[DOI\]](#)[\[Oral\]](#)
- J. Di, K. Hoffmann, T. Chen, **T.-A. Ren**, M. Cutkosky, "SLAP: Slapband-based Autonomous Perching Drone with Failure Recovery for Vertical Tree Trunks", *IEEE Aerospace Conference 2026*. [\[DOI\]](#)
- C. Ng, H. Gao, **T.-A. Ren**, J. Lai, H. Ren, "Contact-Aided Navigation of Flexible Robotic Endoscope Using Deep Reinforcement Learning in Dynamic Stomach", *The 2025 IEEE International Conference on Robotics and Biomimetics (ROBIO 2025)*. [\[Best Paper Award\]](#)[\[DOI\]](#)[\[Oral\]](#)

- C. Xing, H. Li, Y. Wei, **T.-A. Ren**, T. Tu, Y. Lin, E. Schumann, W. Zheng, M. Cutkosky, "TacCap: A Wearable FBG-Based Tactile Sensor for Seamless Human-to-Robot Skill Transfer", *The 2025 IEEE/RSJ International Conference on Intelligent Robots and Systems (IROS 2025)*. [DOI]
- **T.-A. Ren**, W. Liu, T. Zhang, H. Ren, J. Lai, "Three-dimensional Morphological Reconstruction of Millimeter-Scale Soft Continuum Robots based on Dual Stereo Vision", *The 2024 IEEE International Conference on Robotics and Biomimetics (ROBIO 2024)*. [DOI][Oral]
- C. Ng, H. Gao, **T.-A. Ren**, J. Lai, H. Ren, "Navigation of Tendon-Driven Flexible Robotic Endoscope through Deep Reinforcement Learning", 2024. *IEEE International Conference on Advanced Robotics and Its Social Impacts (ARSO 2025)*. [DOI]

Posters & Demonstrations

- **T.-A. Ren**, S. Hong, J. Garcia, J. Grinberg, M. Cutkosky. "PhysioVisionFT: A Multimodal Sensor Framework for Sub-Tissue Feature Detection in Physical Therapy", Bay Area Robotics Symposium 2025.
- **T.-A. Ren**, M. Cutkosky. "Stickybot IV", the Stanford Robotics Center Launch Symposium. [Demo URL]
- **T.-A. Ren**, Z. Yu, B. Vuong, A. Meola, and A. Okamura, "Concentric Agonist-antagonist Robots for Neurosurgical Applications", *IEEE RoboSoft 2024 Workshop*. [URL] [PDF] [Best submission for lighting talk]
- Wang G, **T.-A. Ren**, J. Lai, L. Bai, H. Ren, "Sim-to-Real Segmentation in Robot-assisted Transoral Tracheal Intubation", *IEEE ICRA 2023 Workshop* [DOI] [Best Poster Award]

SERVICES

- **Journal Reviewer:**
 - IEEE Transactions on Medical Robotics and Bionics (T-MRB),
 - IEEE Transactions on Industrial Informatics (T-II),
 - Machine Intelligence Research (MIR).
- **Conference Reviewer:**
 - IEEE International Conference on Intelligent Robots and Systems 2024,
 - IEEE International Conference on Robotics & Automation 2024/2025/2026.

TEACHING & MENTORSHIP EXPERIENCES

- **TA for ME 327: Design and Control of Haptic Systems** (Lecturer: Prof. Allison Okamura) Spring 2025
- **Mentee:** Jorge Garcia (M.S. at Stanford)

HONORS & AWARDS

- **Bio-X Travel Award** at Stanford University 2026
- **Outstanding Graduates in Beijing** 2023
- **National Scholarship of China** 2022
- **Best Paper Award** in *The 2025 IEEE International Conference on Robotics and Biomimetics (ROBIO 2025)*
- **Best Poster Award** in *The 2023 IEEE International Conference on Robotics & Automation Workshop (ICRA 2023)*

SKILLS

- **Programming:** Proficient in C/C++, Python, Assembly Language
- **Software/Platforms:** ROS 1/2, Matlab/Simulink, SOFA Framework, ABAQUS, COMSOL, Multisim, CAD, Gmsh, L^AT_EX, Git, etc.
- **Hardware:** Jetson Nano Developer Kit, STM32, Arduino, NUC, etc.
- **Languages:** Mandarin (Native), English (Proficient)